

Perx Technologies: Reduces data analytics costs by 30%-plus and achieves near 100% availability with Google Cloud

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ABOUT PERX



About Perx Technologies

With Google Cloud
Technologies enable
business leaders
impactful decisions
instantly, by means

Established in 2016 and headquartered in Singapore, [Perx Technologies](https://www.perxtech.com/) (<https://www.perxtech.com/>) provides businesses with a customer data and loyalty platform, delivered as a service, that enables them to improve marketing productivity and build closer relationships with customers. Perx Technologies has grown across South East Asia, with businesses now engaging more than 50 million customers per day on its platform.

consumer engagement behavior in real time

Industries: Technology

Location: Singapore

Google Cloud results

- Achieves 30%-plus reduction in data warehousing and analytics costs
- “Bridges the gap” for customers to measure conversion of digital campaigns to online and physical transactions
- Responds to support queries within one to two hours

Cuts cost
in half
growth
volume

Composer

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Dataprep

(<https://cloud.google.com/dataprep/>)

Operations

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Composer

(<https://cloud.google.com/composer/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

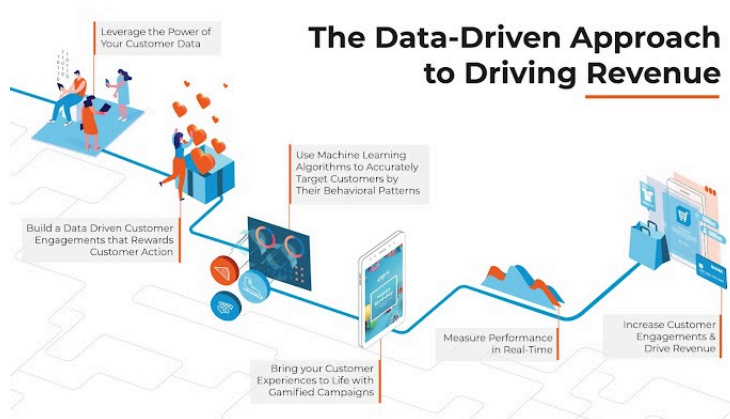
Cloud Storage

(<https://cloud.google.com/storage/>)

Operations

(<https://cloud.google.com/products/operations>)

Securing engagement and loyalty from customers is key to success for many businesses in the digital marketplace. This means interacting with customers over online channels such as websites and offline channels such as physical shopfronts and using data to provide relevant, personalized products and services.



Perx Technologies provides a fully integrated customer data and loyalty software-as-a-service platform. The platform enables organizations of all sizes to monetize customer engagements by creating interactive, revenue-generating digital experiences for millions of end consumers.

For example, rather than simply sending a text message to a customer inviting them to enjoy a discount at select merchants or stores over the Christmas season, B2C organizations can use the Perx platform to provide dynamic, personalized experiences. For example, they may reward customers who maintain a minimum weekly spend

on credit cards with prizes that can be won by participating in games such as digital scratch cards.

With the Perx platform, marketing teams experience a dramatic productivity boost by reducing campaign-to-market timelines from months down to a few minutes.

Businesses engage over 50 million consumers per day on the Perx platform. The platform enables businesses to:

- Improve “last mile” customer engagements—customer actions such as signing up for B2C apps or providing rich personal data by answering survey questions—that drive customer engagement by up to 12x. “Bank promotions typically engage customers at a bare minimum of 2%–5%, depending on the channel used to communicate,” says Yi Ting Ho, Director, Technology Operations, Perx Technologies. “With Perx, average customer engagement levels rise to between 70% and 95%.”
- Drive customers’ propensity to spend on products and services through initiatives such as loyalty programs.
- Boost customer engagement and reduce resistance to cross-sell and upsell initiatives.

Technology strategy key to success

Unsurprisingly, Perx Technologies’ technology strategy is integral to its success. “Our strategy is based on becoming a thought leader in customer loyalty and engagement—pushing boundaries and leveraging data to redefine how businesses can

build and leverage customer loyalty,” explains Rob Roach, VP of Engineering at Perx Technologies. This strategy informs an architecture based on open source technologies and microservices—giving Perx Technologies the ability to integrate seamlessly with external sources such as third-party APIs or imported analytics. Delivering its platform as a service also allows Perx Technologies to scale to meet support traffic bursts during demand peaks and sustain longer-term growth.

Executing this strategy meant using the cloud services that best met Perx Technologies’ needs for specific functions and activities. The organization had been using a query service from another provider, but an analysis of its data warehousing and extract transform load (ETL) requirements found that a data architecture based on the BigQuery (<https://cloud.google.com/bigquery/>) analytics data warehouse was superior to competing offerings.

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BigQuery cost-effective, flexible and intuitive

“BigQuery is cost effective, flexible, intuitive and has more features and third-party integrations than comparative products,” says Roach. The business complemented BigQuery with Dataprep (<https://cloud.google.com/dataprep/>) to explore, clean, and prepare data for analysis; Dataflow (<https://cloud.google.com/dataflow/>) to process data; and Cloud Composer (<https://cloud.google.com/composer/>) workflow orchestration to author and monitor data pipelines across its multi-cloud architecture.

“Google Cloud Singapore provided invaluable support for our implementation by running a detailed education session covering the features, functionality, and usability of BigQuery, Dataprep, Dataflow, and Cloud Composer,” says Roach.

The session kick-started a three-month implementation undertaken by Perx’s DevOps and data teams. The organization ingested application audit logs and implemented change data capture using a data streaming tool to the BigQuery data

warehouse. “Once the data is loaded, we use an open source business intelligence tool to produce data insights, while we use Cloud Composer for ETL, including scheduled reports,” says Roach.

Perx Technologies has expanded its use of Google Cloud services to Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) to run its business intelligence tool and Kafka server in containerized environments; Cloud Storage (<https://cloud.google.com/storage/>) to store data for ETL as required; and Stackdriver (<https://cloud.google.com/stackdriver/>) to provide application monitoring and logging. “We chose Google Kubernetes Engine as it was a mature, managed service and delivered a simplified, efficient deployment process,” says Roach.

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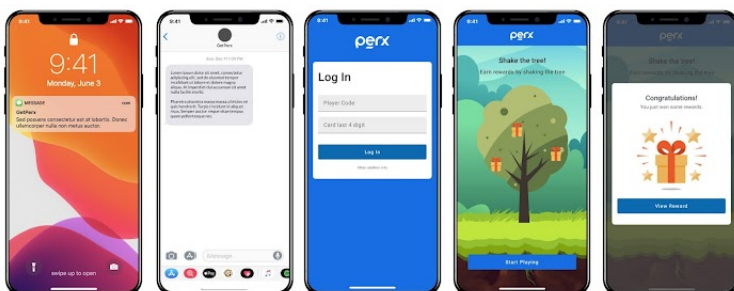
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Query times halved

With BigQuery, Perx Technologies has cut query times in half relative to its previous service—despite the dataset captured in BigQuery rising to more than 5 TB within a year. “Minimal latency and rapid response times are crucial in delivering data to our users in a timely way,” says Roach. “Our average query time is still at lightning speed, despite the large volume of data we have stored.”

“Google Cloud never fails to perform, and we’ve achieve near-100% availability to date.”

The pay-per-use pricing of BigQuery has also proved cost-effective and budget-friendly for Perx Technologies—the business is paying more than 30% less than it would for comparable services from other cloud providers.



Real-time data to monitor campaign performance

With BigQuery, Perx Technologies can now provide real-time data to businesses to track and monitor campaign performance, with meaningful analytics enabling marketers to track “last-mile” engagement—the fulfillment step—with customers. “This bridges the gap for our enterprise clients to better measure conversion of digital campaigns to physical transactions that occur in-store,” says Roach. “The new offering adds greatly to our repertoire of service offerings and enables clients to better understand consumers’ behavior based on engagement histories.”

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With its BigQuery data infrastructure operating smoothly and meeting business needs, Perx Technologies is considering expanding its use of Dataflow to create more data pipelines for its business intelligence tool and transforming data for machine learning. The organization is also exploring the creation of machine learning projects from AI Platform (<https://cloud.google.com/ai-platform/>) to build its recommendation services. “Perx recommendation service is an AI-based tool that will study end-user behavior based on campaign engagement,” says Roach. “With the data we obtain, our engine will propose recommendations to our platform users to better target and engage consumers.”

“Adding Google Cloud to our environment has been a breeze—the support team has been incredibly helpful with an extremely fast turnaround in responses to queries we’ve made to the online help desk,” concludes Roach. “When it comes to Google Cloud services, maintenance and scaling has been hassle-free, while we love the pay-per-use pricing structure.”

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Reach out to
our team to
see how
Google Cloud
can help your

business.